Your Diversity Metric Is Measuring Item Difficulty

Abstract

Claims that language models "fail on the same inputs" are usually supported by comparing a pairwise agreement or co-failure rate against an independence baseline. We show that comparison is uninformative by construction. Mean pairwise co-failure is exactly the *double-fault* diversity measure of the classifier-ensemble literature [Kuncheva & Whitaker, 2003], and it is a deterministic function of the item margins, so its excess over any item-margin-preserving null is identically zero — verified to 10^{-16} on five benchmarks. We apply the exact conditional null implied by Rasch sufficiency, following standard nonparametric Rasch methodology [Ponocny, 2001], to 1,228–1,373 open models per benchmark, and — unlike prior uses — we characterise the test's power rather than asserting its validity. Three results follow. The residual correlation that survives conditioning is 2.9–11.9× its null and is *not* a duplicate-model artifact: removing every model agreeing with a retained model above 0.95 discards 36–64% of each population and moves 14 of 15 headline statistics by less than 1.5×. The test has power 1.00 against planted shared failure modes at a strength well below what is observed, and detects copying from a rate of 0.1. And it has power exactly α against the alternative in which all models share one item-difficulty profile — that alternative *is* the null, so the strongest form of monoculture is invisible to this design by construction. We state that blind spot as a property, not a caveat. We also show that the statistic most tempting to report, an "effective number of independent models," is a monotone function of mean squared residual correlation and counts nothing; the same objection applies to the Kish design effect used in recent LLM-judge work [Kohli, 2026].

1 The degenerate statistic

Let $F \in \{0,1\}^{N\times M}$ with $F_{im} = 1$ iff model i fails item m; write $c_m = \sum_i F_{im}$ and $O = \frac{1}{MN(N-1)} \sum_{i\neq j} \sum_m F_{im}F_{jm}$ for mean pairwise co-failure.

Lemma 1. $O = \dfrac{1}{MN(N-1)} \displaystyle\sum_m \big(c_m^2 - c_m\big).$

Proof. $\sum_{i\neq j} F_{im}F_{jm} = (\sum_i F_{im})^2 - \sum_i F_{im}^2 = c_m^2 - c_m$ since $F_{im} \in \{0,1\}$. □

O depends on the item margins alone. Any randomisation preserving them leaves O exactly invariant, so its "excess over independence" is identically zero and its standardised effect size is $0/0$. The scope matters: this follows from preserving *item* margins, so it does not by itself impugn analyses whose baseline conditions on model accuracy.

O is not an unfamiliar quantity in machine learning — it is the double-fault measure, one of the ten catalogued by Kuncheva & Whitaker [2003]. The degeneracy is visible in that literature's own regression design: over 5,999 panels sampled from our ARC response data, mean member accuracy and panel size alone explain $R^2 = 0.9928$ of majority-vote accuracy, and the standardised coefficients on member accuracy and double-fault diverge to +316 and +275 — the signature of the collinearity Lemma 1 predicts. Adding both classical diversity measures gains $\Delta R^2 = 0.0017$; adding a difficulty-conditioned measure gains 0.0002, with the wrong sign. **Diversity metrics, corrected or not, do not help panel construction at this scale.** This converges with independent, contemporaneous work: Kim [2026] audits five diversity measures against realised majority-vote gain over 31,900 LLM-ensemble subsets and shows the same statistics are algebraically rank-deficient once mean accuracy is controlled, by a different route (no margin-preserving null, gain-over-best-member as target) — we flag the overlap rather than claim the finding as ours alone.

The degeneracy has been proved independently in three literatures — as Schluter's V-ratio in ecology, Kuder–Richardson/Cronbach's α in psychometrics, and Fleiss's observed agreement for the multi-category case. We claim the translation, not the result.

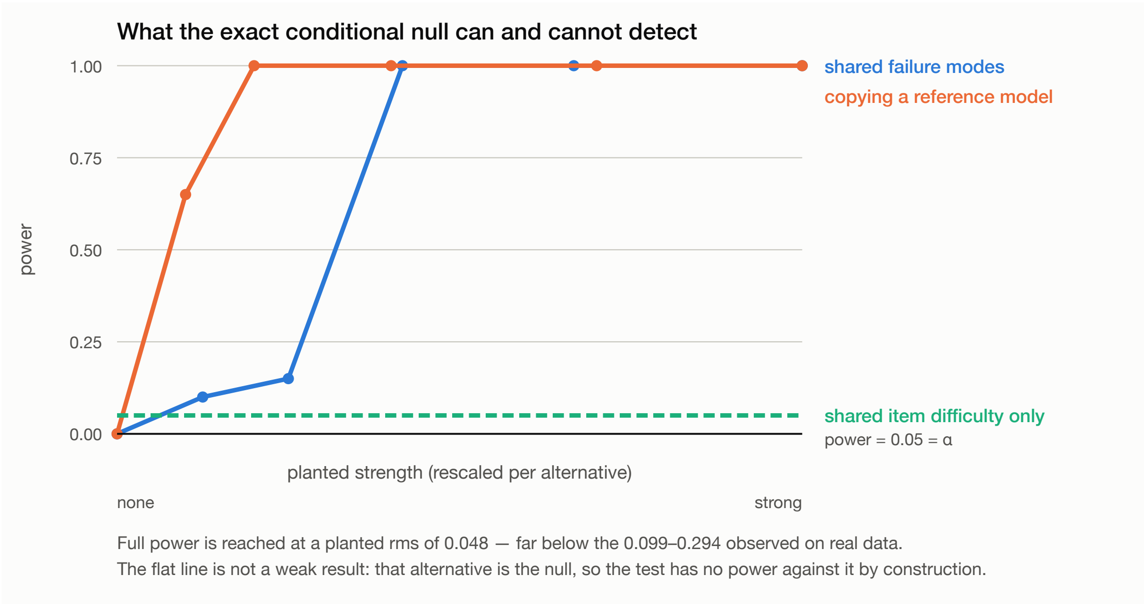

Figure 1: Detection power against three planted alternatives. The flat line is the design's blind spot, not a weak result: models sharing one item-difficulty profile and differing only in ability are exactly what the null describes.

2 The null, and what it can see

Row and column margins are the jointly sufficient statistics of the Rasch family, so the uniform distribution over margin-matched matrices is an exact, fit-free conditional null for that family. This is standard nonparametric Rasch methodology [Ponocny, 2001, Verhelst, 2008]; our contribution is to apply it at this scale and to characterise it. We sample it with curveball [Strona et al., 2014], which is uniform on the fibre provided failed trades are counted [Carstens, 2015].

The sampler is uniform (verified). Enumerating every binary matrix with given margins on four fibres (sizes 48, 90, 204, 1170) and drawing 60,000 thinned samples, a χ^2 test against uniformity is not rejected on any ($p = 0.05, 0.23, 0.73, 0.15$). Four dispersed chains give Gelman–Rubin $\hat{R} \in [0.98, 1.03]$ on all five benchmarks.

The null is not the noise floor. The exact null's residual-correlation rms sits at 1.25–2.01× the analytic $1/\sqrt{M-3}$ floor. Two rungs are informative: conditioning on model margins alone lands exactly *on* that floor, while conditioning on item margins alone reproduces the both-margin null to within 6%. Empirically the item margins do essentially all the conditioning work.

Power, and the blind spot. On planted data (400 models, 700 items, 20 datasets per condition, $\alpha = 0.05$; null conditions reject at 0.017):

planted alternative	strength	power
shared failure modes	$s = 0.3$ / $s = 0.5$	0.15–0.25 / **1.00** (at rms 0.048)
copying a reference model	$c = 0.05$ / $c = 0.10$	0.65 / **1.00**
shared item difficulty only	—	**0.05** $= \alpha$

Full power is reached at a planted rms of 0.048, far below the 0.099–0.294 observed (Figure 1). The third row is the Narcissus effect of Colwell & Winkler [1984] in exact form: if every model finds the same items hard and differs only in ability, that case *is* the null and the test cannot see it. A null result here is not evidence of independence. We report this in the abstract because a reader who does not know it will misread a negative finding.

3 What survives, on 1,228–1,373 models per benchmark

Substrate: five benchmarks from the Open LLM Leaderboard v1 archive (July 2023–June 2024), harvested at zero compute cost by column-selective parquet reads. Residual correlation is the rms off-diagonal of the margin-conditioned excess matrix.

	rms residual correlation		after dedup at 0.95		dims
benchmark	observed	vs. null	% removed	ratio	above edge
ARC-Challenge	0.2483	5.58×	36%	5.44×	6
Winogrande	0.2691	6.58×	34%	6.18×	5
TruthfulQA	0.2942	5.06×	40%	4.64×	5
GSM8K	0.0987	2.85×	21%	3.10×	4
HellaSwag	0.2457	11.84×	64%	13.45×	12

The effect is not duplication. This is the objection the population invites, and it does not hold for the population-level statistics: 14 of 15 statistic–benchmark pairs move by less than 1.5× between the full and 0.95-deduplicated populations (the exception is HellaSwag's participation ratio, 2.44×). It *does* hold for a gradient claim we therefore do not make: apparent redundancy concentrating among the highest-accuracy models collapses under the same deduplication on three of five benchmarks.

Dimensionality. Counting eigenvalues above the null spectral edge is Horn's parallel analysis with a stronger reference than Horn's own — the exact margin-preserving null already carries the observed ability and difficulty profiles. Retained counts are 4–6 (12 on HellaSwag), stable to within one across deduplication thresholds on four of five benchmarks. These are upper bounds; eigenvalue-counting rules over-retain. This bears directly on IRT-based benchmark compression [Polo et al., 2024], which fits latent-ability models to matrices of this kind — using 319 models, roughly a quarter of the population here — and rests on an assumed dimensionality. Our test needs no IRT fit of its own, so it can check that assumption.

4 What we do not claim

We do *not* report an "effective number of independent models." The participation ratio is algebraically $N/(1+(N-1)\overline{R_{ij}^2})$, a monotone function of the mean squared residual correlation, so it cannot distinguish one weak global factor from many tight clusters. The same objection applies to the Kish design effect $k/(1+(k-1)\bar{\phi})$ used as the headline statistic in recent LLM-judge panel work [Kohli, 2026], differing only in using the mean rather than the mean square. Sha & Zhao [2026] apply the identical participation-ratio statistic on the orthogonal axis — counting effective independent *benchmarks* rather than models — and reach the same conclusion by a different argument, explicitly treating it as a screening diagnostic rather than a literal factor count.

The central limitation is identification: our conditioning family is unidimensional, and we cannot separate genuine behavioural concentration from the Rasch family being too small. A pre-registered test on real responses found that our own critique does *not* rescue a specific published claim — the accuracy–agreement association attenuates only 12% under conditioning — and we report that prior finding as robust. A pre-registered dispersion statistic was refuted by its own kill condition, and a lineage decomposition was dropped when its coverage gate failed at 23.3% against a 40% threshold. All code, substrate, results artifacts and the pre-registration with dated amendments are released; every number above traces to an executed run.

Use of AI assistance. A large language model was used as a coding and analysis assistant. The author directed the research, pre-registered the hypotheses and kill conditions, verified the derivations, and takes full responsibility, in line with COPE guidance that AI tools cannot be authors.

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